Jeongjin Han

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Research Interests

- Memory-centric systems and data movement optimization
- Large-scale simulation, parallelism, and performance optimization
- System state management, checkpointing, and deterministic execution

Education

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, Korea
B.S. in Computer Science & Nuclear and Quantum Engineering — Feb. 2022 – Present

- **GPA:** 3.92 / 4.3 (96.20 / 100)
- **Minor:** Mathematical Sciences
- **Honors:** Dean's List, Spring 2025

Research Experience

Reactor Physics and Transmutation Lab, KAIST — Winter 2024
Undergraduate Researcher | Advisor: Prof. Yonghee Kim

- Conducted Monte Carlo neutron transport simulations using **OpenMC**, analyzing neutron moderation behavior under Thermal Scattering Law (TSL) models from the ENDF/B-VIII.1 nuclear data library.
- Investigated computational and memory characteristics of large-scale Monte Carlo simulation workloads, and analyzed scalability behavior under parallel execution.

Nuclear I&C and Autonomous Operation Lab, KAIST — Winter 2025
Undergraduate Researcher | Advisor: Prof. Jonghyun Kim

- Identified GPU underutilization bottleneck caused by a CPU-bound simulator; redesigned the training pipeline to decouple policy optimization from environment interaction, improving hardware utilization under system resource constraints.
- Reverse-engineered the process memory layout of a closed-source Windows VM simulator to extract and inject runtime state variables, enabling system integration without source access or binary modification.
- Designed and implemented a TCP/Docker-based distributed execution infrastructure bridging a Windows VM and a GPU server, managing cross-system state synchronization and execution control.
- Enabled efficient training of RL agents under system-level constraints including memory bandwidth and execution latency bottlenecks.

Projects

KECC: Optimizing C-to-RISC-V Compiler — Fall 2025
CS420 Compiler Design

- Designed and optimized a C-to-RISC-V compiler in **Rust** covering the full pipeline: IR generation, SSA-based optimization passes (memory-to-register promotion via Mem2reg, GVN, CFG simplification, dead-code elimination) to reduce memory traffic and instruction count, and a RISC-V Asmgen backend with full struct/array/float/pointer support and SSA destruction via phi-node parallel-move insertion.
- Diagnosed and patched 10+ correctness bugs across all phases (non-deterministic GVN hash collisions, erroneous dead GEP elimination, float calling-convention violations, wrong-block phi-argument removal, etc.).
- Evaluated generated code on **gem5 v25.1 MinorCPU** (cycle-accurate in-order microarchitecture) across 60 C benchmarks: geometric-mean cycle ratio **1.009**× vs. Clang -O1, with 37/60 programs outperforming Clang.

`Jeong-jin-Han/CS420-KECC`

Nuclear System Code: Overcooling Accident Simulation — Fall 2025
NQE625 Nuclear System Computational Analysis

- Implemented a multi-channel 1-D nuclear system code from scratch in **MATLAB** modeling cylindrical geometry with axial discretization (z-direction neglected); applied upwind-scheme finite-volume discretization and semi-implicit time integration across coupled subsystems: fluid momentum/enthalpy transport, point kinetics with delayed neutrons, fuel heat conduction (FVM), and steam generator primary–secondary coupling.
- Performed steady-state and transient analyses of an overcooling accident; validated numerical solutions against the industry-standard **MARS** code across 10 physical quantities, demonstrating mathematical applicability of numerical methods to large-scale coupled system simulation.

VRAIL: Vectorized Reward-based Attribution for Interpretable Learning Fall 2025
CS377 Reinforcement Learning

- Proposed a bi-level RL framework learning interpretable linear/quadratic value estimators as potential-based reward shaping functions, improving DQN convergence stability in Taxi-v3; contributed reward transfer experiments and α-scheduling trials.

`arXiv:2506.16014`

SHIFT: Sigmoid-Based Heuristic Invertible Fitness-Landscape Transformation Fall 2025
CS454 AI-Based Software Engineering

- Designed a sigmoid-based invertible fitness-landscape transformation for SBST, enabling hill climbing to escape plateaus; applied compiler techniques (AST-based instrumentation, constant folding, SSA-like variable-shading) to Python source transformation, and implemented parallel time-budgeted evaluation infrastructure.

`Jeong-jin-Han/SHIFT-SBST`

Skills

Languages	C++, Rust, Python, MATLAB, Scala
Tools & Frameworks	PyTorch, gem5, Docker, HuggingFace, LaTeX, Git, Linux

Relevant Coursework

Systems	Computer Organization (CS.30101) Operating Systems and Lab (CS.30300) System Programming (CS.20300) Introduction to Computer Networks (CS.30401)
CS Theory	Introduction to Algorithms (CS.30000) Data Structure (CS.20006) Compiler Design (CS.40200) Programming Language (CS.30200)
AI / ML	Introduction to Reinforcement Learning (CS.30707) Machine Learning (CS.30706) Introduction to Artificial Intelligence (CS.40700)
Numerical / HPC	Numerical Methods and Computer Simulation (NQE.30011) Computational Analysis in Nuclear System (NQE.60025)
Mathematics	Differential Equations and Applications (MAS.20001) Analysis I / II (MAS.20041/42) Lebesgue Integral Theory (MAS.40041) Probability & Introductory Random Processes (EE.20010)